BCT Practical 2 – Create Your Own Wallet Using MetaMask for Crypto Transactions

PART 1 – UNDERSTANDING THE BASIC CONCEPTS

1. What is Blockchain?

Blockchain is a digital record-keeping system that stores information in blocks connected together in a chain.

It is a decentralized system, which means that data is maintained across multiple computers instead of being controlled by one central authority.

How does blockchain work?

1. A user initiates a transaction.
2. The transaction is broadcast to the blockchain network.
3. Network participants validate the transaction.
4. Valid transactions are grouped into blocks.
5. The block is added to the existing blockchain.
6. The transaction becomes part of the blockchain's permanent record.

Example

Suppose Person A sends 1 ETH to Person B.

- Person A initiates the transaction.
- The network verifies it.
- The transaction is recorded on the Ethereum blockchain.
- Person B receives the ETH.

Features of Blockchain

- **Decentralization:** No single central authority controls the entire network.
- **Transparency:** Transactions on public blockchains can be viewed by anyone.
- **Security:** Cryptography helps protect transactions and accounts.
- **Immutability:** Confirmed blockchain records are designed to be extremely difficult to change.

- **Distributed ledger:** Multiple computers maintain copies of the blockchain data.

2. What is Cryptocurrency?

Cryptocurrency is a digital form of currency that uses cryptographic techniques to secure transactions.

Unlike traditional currency, cryptocurrencies can be transferred through blockchain networks without requiring a traditional bank to process every transaction.

Examples:

- Bitcoin (BTC)
- Ethereum's Ether (ETH)
- Litecoin (LTC)

3. What is Ethereum?

Ethereum is a decentralized blockchain platform that allows users to transfer digital assets and execute smart contracts.

It supports:

- Cryptocurrency transactions.
- Smart contracts.
- Decentralized applications (DApps).
- Decentralized finance (DeFi).

Example: In our BCT practicals, we use Ethereum to deploy the BankAccount and StudentRegister smart contracts.

4. What is Ether (ETH)?

Ether, commonly called ETH, is the native cryptocurrency of the Ethereum blockchain.

It is used for:

1. Sending and receiving digital currency.
2. Paying transaction fees.
3. Deploying smart contracts.
4. Interacting with decentralized applications.

Example: When we deploy our BankAccount smart contract through Remix, we pay a small amount of Sepolia ETH as a test transaction fee.

Difference between Ethereum and Ether

Ethereum	Ether
A blockchain platform.	Its native cryptocurrency.
Supports smart contracts and DApps.	Used for transactions and gas fees.
It is the network.	It is the digital currency.

Remember: Ethereum is the platform, and Ether is the currency used on that platform.

5. What is a Cryptocurrency Wallet?

A cryptocurrency wallet is a digital application that allows users to manage blockchain accounts and interact with cryptocurrency networks.

It does not actually store cryptocurrency like a physical wallet stores cash.

Instead, it manages cryptographic keys that allow users to access and control their digital assets recorded on the blockchain.

A wallet allows users to:

- Send cryptocurrency.
- Receive cryptocurrency.
- Check account balances.
- Sign and approve transactions.
- Connect to decentralized applications.

Examples:

- MetaMask
- Trust Wallet
- Coinbase Wallet

6. What is MetaMask?

MetaMask is a cryptocurrency wallet that allows users to interact with Ethereum and other compatible blockchain networks.

It is available as a browser extension and a mobile application.

Main uses of MetaMask

1. Create and manage cryptocurrency wallets.
2. Send and receive ETH and supported tokens.
3. Connect to decentralized applications.
4. Approve blockchain transactions.

5. Interact with smart contracts.
6. Switch between supported blockchain networks.

Why are we using MetaMask in our practical?

We use MetaMask to create our wallet, connect it to Remix IDE and approve transactions when deploying and interacting with smart contracts.

7. Important Components of MetaMask Wallet

A. Public Address

A public address is a unique address associated with a blockchain account.

- It is used to receive cryptocurrency.
- It can be shared with other people.
- It usually starts with `0x` on Ethereum.

Example:

```
0x1234567890abcdef...
```

This is only an illustrative address, not a real wallet.

B. Private Key

A private key is a secret cryptographic key associated with an account.

- It is used to digitally sign transactions.
- It proves control over the account.
- It must never be shared with anyone.

Simple example: A public address is like your bank account number, whereas the private key is like the secret authorization that lets you control the account.

C. Secret Recovery Phrase (SRP)

A Secret Recovery Phrase is generally a sequence of 12 words generated when creating a traditional MetaMask wallet.

It is used to recover the wallet if access is lost.

Important points:

- It is the master recovery key for accounts derived from that wallet.
- Anyone who obtains it can gain control of the associated accounts.
- It must be stored securely.
- It should never be shared or entered on unknown websites.

D. Wallet Password

The password protects access to MetaMask on your device.

It is different from the Secret Recovery Phrase.

Password	Secret Recovery Phrase
Used to unlock MetaMask on a device.	Used to restore the wallet.
Can be changed.	Generated during wallet creation and cannot be edited.
Does not replace the recovery phrase.	Provides access to the wallet and its derived accounts.

Note: MetaMask also offers social-login setup in supported versions. For our traditional wallet-creation procedure, we use the Secret Recovery Phrase method.

8. What is a Blockchain Network?

A blockchain network is a group of computers that communicate with each other to maintain and validate blockchain transactions.

MetaMask allows users to connect to different networks.

Important Ethereum networks:

A. Ethereum Mainnet

- The actual Ethereum production network.
- Real ETH is used.
- Transactions involve real financial value.
- Smart contracts deployed here are accessible on the production network.

B. Ethereum Sepolia Testnet

- A test network used by developers.
- Used to test smart contracts and applications.
- Uses test ETH, which has no real-world monetary value.
- Helps developers experiment without spending real ETH.

Why do we use Sepolia?

We use Sepolia because our practicals involve deploying and testing smart contracts. It allows us to perform transactions without using real money.

9. What is a Faucet?

A faucet is a website or service that provides free test cryptocurrency to users.

It is used to obtain test ETH for blockchain development and experimentation.

Examples:

- Google Cloud Web3 Faucet.
- Alchemy Sepolia Faucet.
- QuickNode Faucet.
- Sepolia PoW Faucet.

Example: We copy our MetaMask wallet address and paste it into a Sepolia faucet. The faucet sends test ETH to that address.

10. What is a Transaction?

A transaction is an operation performed on a blockchain.

Examples:

- Sending ETH to another wallet.
- Deploying a smart contract.
- Calling a function that changes smart contract data.

A transaction generally contains:

- Sender address.
- Recipient address or contract address.
- Value (if Ether is transferred).
- Transaction data.
- Gas-related information.
- Digital signature.

After submission, the network processes the transaction and records its result.

11. What is Gas?

Gas is the unit used to measure the computational work required to perform operations on Ethereum.

For example:

- Sending ETH requires gas.
- Deploying a smart contract requires gas.
- Updating smart contract data requires gas.

Gas fees are paid in ETH.

Important gas terms

Term	Meaning
Gas	Unit of computational work.
Gas Used	Actual gas consumed.
Gas Limit	Maximum gas allocated for a transaction.
Gas Price	Cost per unit of gas.
Transaction Fee	Total fee paid for the transaction.

Formula:

Transaction Fee = Gas Used × Effective Gas Price

12. What are Wei and Gwei?

Ether has smaller denominations used to represent very small amounts.

- Wei is the smallest unit of Ether.
- Gwei is commonly used to express gas prices.

Conversions:

1 ETH = 1,000,000,000,000,000,000 Wei

1 ETH = 1,000,000,000 Gwei

1 Gwei = 1,000,000,000 Wei

Example:

0.001 ETH = 1,000,000,000,000,000 Wei.

This is why we sometimes enter large numbers in Remix when specifying Ether amounts in Wei.

13. What is a Smart Contract?

A smart contract is a program deployed on a blockchain that executes predefined instructions when called.

It can:

- Store data.
- Transfer digital assets.
- Validate conditions.
- Execute predefined business logic.

Example: In our BankAccount practical, the smart contract allows users to deposit Ether, withdraw Ether and check balances.

14. What is Remix IDE?

Remix is an online development environment used to write, compile, deploy and test Solidity smart contracts.

Website: https://remix.ethereum.org/

We use Remix in our subsequent practicals to:

1. Write Solidity programs.
2. Compile smart contracts.
3. Connect MetaMask.
4. Deploy contracts on Sepolia.
5. Execute smart contract functions.

15. What is Etherscan?

Etherscan is a blockchain explorer used to view and verify Ethereum blockchain information.

For Sepolia, we use:

https://sepolia.etherscan.io/

It allows us to:

- Search wallet addresses.
- View transactions.
- Check transaction status.
- View gas used and transaction fees.
- View deployed smart contracts.

PART 2 – PERFORMING THE PRACTICAL STEP BY STEP

Aim

To create a cryptocurrency wallet using MetaMask for crypto transactions and configure it to interact with the Ethereum blockchain.

Requirements

- Laptop or desktop computer.
- Internet connection.
- Google Chrome or Microsoft Edge browser.
- MetaMask browser extension.
- Sepolia Testnet.
- Free test ETH from a faucet.

Step 1 – Download and Install MetaMask

<AsyncImage query="MetaMask official website download browser extension homepage" aspectRatio="16:9" width="100%"/>

1. Open Google Chrome or Microsoft Edge.
2. Visit the official website: https://metamask.io/
3. Click on **Download**.
4. Select your browser.
5. You will be redirected to the official browser extension store.
6. Click **Add to Chrome** or the equivalent installation option.
7. Wait for the extension to install.
8. Pin the MetaMask extension to your browser toolbar for easy access.

Why are we doing this?

We are installing MetaMask because it will act as our cryptocurrency wallet and allow us to interact with the Ethereum blockchain.

Important: Install MetaMask only through its official website and linked extension store.

Step 2 – Create a New Wallet

<AsyncImage query="MetaMask extension create new wallet welcome onboarding screen" aspectRatio="16:9" width="100%"/>

1. Click the MetaMask extension icon.
2. Select **Create a new wallet**.
3. Choose the Secret Recovery Phrase setup option if presented.
4. Read and accept the required terms.
5. Create a strong password.

6. Confirm the password.
7. Continue to the wallet recovery setup.

Why do we create a password?

The password protects access to the wallet on our device.

Step 3 – Save the Secret Recovery Phrase

<AsyncImage query="MetaMask secret recovery phrase reveal wallet setup screen" aspectRatio="16:9" width="100%"/>

1. MetaMask generates a Secret Recovery Phrase.
2. Reveal the phrase when prompted.
3. Write down the words in their exact order.
4. Store them securely offline.
5. Complete the verification by selecting the words in the correct order.
6. Finish the wallet setup.

Why is this important?

The recovery phrase is used to restore the wallet if we lose access to our device or MetaMask installation.

Security precautions:

- Never share your recovery phrase.
- Never send it to anyone through WhatsApp, email or other platforms.
- Never enter it on an unknown website.
- MetaMask support will never need your recovery phrase.

Step 4 – View Your Wallet Address

<AsyncImage query="MetaMask wallet account homepage copy public account address" aspectRatio="16:9" width="100%"/>

1. Open MetaMask.
2. Select your account.
3. Find the wallet address displayed below the account name.
4. Click the address to copy it.
5. The address is your public Ethereum account address.

Why do we need this address?

We use this address to receive test ETH from faucets and to identify our account on the blockchain.

Remember: The public address can be shared, but the private key and recovery phrase must remain secret.

Step 5 – Enable Sepolia Test Network

<AsyncImage query="MetaMask wallet network settings show test networks Sepolia selection" aspectRatio="16:9" width="100%"/>

MetaMask may hide test networks by default.

1. Open MetaMask.
2. Open the network selection menu or network settings.
3. Find the option to show test networks.
4. Enable **Show test networks**.
5. Return to the network selection menu.
6. Select **Sepolia**.

Now your MetaMask wallet is connected to the Ethereum Sepolia Test Network.

Why do we select Sepolia?

Because we want to perform practical transactions and deploy smart contracts using test ETH instead of real ETH.

Important: Always check the selected network before sending transactions or deploying contracts.

Step 6 – Get Free Sepolia ETH Using a Faucet

<AsyncImage query="Google Cloud Web3 Ethereum Sepolia faucet request test ETH page" aspectRatio="16:9" width="100%"/>

A faucet provides free test ETH to developers.

We can use the Google Cloud Web3 Faucet:

https://cloud.google.com/application/web3/faucet/ethereum/sepolia

Procedure:

1. Open the faucet website.
2. Copy your MetaMask public address.
3. Paste the address into the faucet input field.
4. Complete any required verification.
5. Click the request button.
6. Wait for the request to be processed.
7. Open MetaMask.
8. Make sure Sepolia is selected.

9. Check whether your test ETH balance has increased.

Other faucet options:

- Alchemy: https://www.alchemy.com/faucets/ethereum-sepolia
- QuickNode: https://faucet.quicknode.com/ethereum/sepolia
- Sepolia PoW Faucet: https://sepolia-faucet.pk910.de/

Why do we need test ETH?

We need test ETH to pay the gas fees for deploying and interacting with smart contracts on Sepolia.

Remember: Test ETH has no real monetary value.

Step 7 – Understand How to Receive ETH

Suppose your friend wants to send you test ETH.

1. Open MetaMask.
2. Copy your public wallet address.
3. Share the address with your friend.
4. Your friend selects the Sepolia network and sends test ETH to that address.
5. Wait for the transaction to be confirmed.
6. Check your MetaMask balance.

The sender needs to pay the transaction fee.

Step 8 – Understand How to Send ETH

For learning purposes, you can perform a small Sepolia test transaction to another wallet.

1. Open MetaMask.
2. Confirm that Sepolia is selected.
3. Click **Send**.
4. Enter the receiver's Sepolia wallet address.
5. Enter a small test ETH amount.
6. Review the transaction details and estimated gas fee.
7. Click Confirm.
8. Wait for the transaction to be processed.
9. Check the transaction status in MetaMask.

Important: Never use your real ETH or mainnet for this practice exercise. Double-check the receiver's address and selected network.

Step 9 – Verify the Transaction on Etherscan

<AsyncImage query="Sepolia Etherscan transaction details overview status gas used transaction fee" aspectRatio="16:9" width="100%"/>

1. Open https://sepolia.etherscan.io/.
2. Copy the transaction hash from MetaMask activity.
3. Paste it into the Etherscan search bar.
4. Open the transaction details.
5. Observe:
    - Transaction status.
    - From address.
    - To address.
    - Value transferred.
    - Gas used.
    - Transaction fee.

This helps us understand how transactions are recorded on the blockchain.

PART 3 – HOW WE USE THIS WALLET IN OUR OTHER PRACTICALS

This is very important because Practical 2 is the foundation for Practical 3 and Practical 4.

Connecting MetaMask with Remix IDE

<AsyncImage query="Remix IDE deploy and run transactions environment browser extension MetaMask" aspectRatio="16:9" width="100%"/>

1. Open https://remix.ethereum.org/.
2. Open the Solidity file containing your smart contract.
3. Compile the contract.
4. Open the **Deploy & Run Transactions** tab.
5. Select **Remix Browser Extension** as the environment.
6. Select your MetaMask account when prompted.
7. Approve the connection.
8. Ensure MetaMask is on Sepolia.
9. Select the contract and click Deploy.
10. Approve the transaction in MetaMask.
11. Wait for successful deployment.

What happens when we click Deploy?

- Remix sends the deployment request.
- MetaMask displays the transaction details.

- We approve the transaction.
- The transaction is broadcast to Sepolia.
- The network executes the deployment.
- The deployed contract receives its own blockchain address.

How do we use MetaMask in Practical 3?

In the BankAccount practical:

- We used MetaMask to connect Remix.
- We deployed the BankAccount contract.
- We deposited 0.001 ETH.
- We withdrew 0.0004 ETH.
- We checked the remaining balance.

How do we use MetaMask in Practical 4?

In the StudentRegister practical:

- We deployed StudentRegister and TestPayable.
- We registered student details.
- We retrieved student information.
- We tested fallback and receive functions.
- We observed gas consumption and transaction fees.

PART 4 – IMPORTANT DIFFERENCES

1. Wallet vs Account

Wallet	Account
Software used to manage keys and interact with blockchain networks.	An identity on the blockchain with an address and associated keys.
Example: MetaMask.	Example: Your Ethereum account.

2. Mainnet vs Testnet

Mainnet	Testnet
Actual production blockchain.	Testing blockchain.
Uses real ETH.	Uses test ETH.

Transactions have real financial value.	Test ETH has no real monetary value.
Used for actual applications.	Used for development and testing.

3. Public Key vs Private Key vs Public Address

Term	Meaning
Public Key	Cryptographic key associated with an account.
Private Key	Secret key used to sign transactions.
Public Address	Address derived from the public key and used to receive assets.

4. ETH vs Wei vs Gwei

Unit	Conversion
ETH	Main currency unit
Gwei	1 ETH = 1 billion Gwei
Wei	1 ETH = 10^{18} Wei

5. MetaMask vs Remix vs Etherscan

Tool	Purpose
MetaMask	Manages wallet and approves transactions.
Remix IDE	Writes, compiles, deploys and tests smart contracts.
Etherscan	Views and verifies blockchain transactions.
Faucet	Provides test ETH.

PART 5 – COMMON ERRORS AND SOLUTIONS

Problem	Solution

MetaMask extension not opening	Check whether the extension is installed and enabled.
Sepolia not visible	Enable Show test networks in MetaMask.
Faucet not sending ETH	Check eligibility, faucet limits and network selection.
ETH not visible	Make sure Sepolia is selected and wait for transaction confirmation.
Remix not connecting	Select Remix Browser Extension and approve the connection in MetaMask.
Transaction likely to fail	Check the function inputs, selected account, network and contract conditions before confirming.
Insufficient funds	Obtain more Sepolia test ETH from a faucet.
Wrong network	Switch MetaMask to Sepolia before performing the practical.

PART 6 – IMPORTANT VIVA QUESTIONS AND ANSWERS

Basic Concepts

Q1. What is blockchain?

Blockchain is a decentralized digital ledger that stores information in connected blocks across multiple computers.

Q2. What is cryptocurrency?

Cryptocurrency is a digital currency that uses cryptography to secure transactions.

Q3. What is Ethereum?

Ethereum is a decentralized blockchain platform that supports cryptocurrency transactions and smart contracts.

Q4. What is Ether?

Ether or ETH is the native cryptocurrency of the Ethereum blockchain. It is used for transactions and gas fees.

Q5. What is the difference between Ethereum and Ether?

Ethereum is the blockchain platform, whereas Ether is the cryptocurrency used on that platform.

Q6. What is a cryptocurrency wallet?

A cryptocurrency wallet is a digital application that manages cryptographic keys and allows users to send, receive and manage digital assets.

MetaMask Questions

Q7. What is MetaMask?

MetaMask is a cryptocurrency wallet used to manage blockchain accounts and interact with Ethereum-based applications.

Q8. Why do we use MetaMask?

We use MetaMask to create wallets, send and receive cryptocurrency, connect to Remix and approve smart contract transactions.

Q9. What is a public address?

A public address is a unique blockchain address used to receive cryptocurrency.

Q10. What is a private key?

A private key is a secret cryptographic key used to sign and authorize transactions.

Q11. What is a Secret Recovery Phrase?

It is a sequence of words used to recover a cryptocurrency wallet.

Q12. Can we share our public address?

Yes. It is used to receive cryptocurrency and can be shared publicly.

Q13. Can we share our private key?

No. Anyone with the private key can control the associated account.

Q14. What happens if we lose our Secret Recovery Phrase?

If we lose access to our wallet and its recovery method, we may permanently lose access to our assets. MetaMask cannot recover the phrase for us.

Q15. What is the difference between a password and a recovery phrase?

The password unlocks MetaMask on a device, whereas the recovery phrase restores the wallet.

Ethereum Network Questions

Q16. What is a blockchain network?

It is a group of connected computers that maintain and validate blockchain transactions.

Q17. What is Ethereum Mainnet?

It is the actual production network of Ethereum where real ETH is used.

Q18. What is a testnet?

A testnet is a blockchain network used for testing applications without using real cryptocurrency.

Q19. What is Sepolia?

Sepolia is an Ethereum test network used mainly for testing decentralized applications and smart contracts.

Q20. Why do we use Sepolia instead of Mainnet?

Because Sepolia allows us to test and deploy smart contracts using free test ETH without spending real money.

Q21. What is a faucet?

A faucet is a service that provides free test cryptocurrency to users.

Q22. Does Sepolia ETH have real value?

No. Sepolia ETH is used only for testing and has no real-world monetary value.

Transaction and Gas Questions

Q23. What is a blockchain transaction?

It is an operation performed on a blockchain, such as transferring ETH or interacting with a smart contract.

Q24. What is gas?

Gas is the unit used to measure the computational work required to execute operations on Ethereum.

Q25. Why do we pay gas fees?

Gas fees compensate the network for processing transactions and help prevent spam.

Q26. What is Gas Used?

It is the actual amount of gas consumed during transaction execution.

Q27. What is Gas Limit?

It is the maximum amount of gas allocated for a transaction.

Q28. What is Wei?

Wei is the smallest denomination of Ether.

Q29. What is Gwei?

Gwei is a denomination of Ether commonly used to express gas prices.

Q30. What is Etherscan?

Etherscan is a blockchain explorer used to view and verify Ethereum transactions, addresses and smart contracts.

Smart Contract and Development Questions

Q31. What is a smart contract?

A smart contract is a program deployed on a blockchain that executes predefined instructions.

Q32. What is Solidity?

Solidity is a programming language used to develop smart contracts for Ethereum-compatible blockchains.

Q33. What is Remix IDE?

Remix is an online development environment used to write, compile, deploy and test Solidity smart contracts.

Q34. Why do we connect MetaMask with Remix?

We connect MetaMask to Remix to select an account, approve transactions and deploy smart contracts on the blockchain.

Q35. What is a DApp?

A decentralized application is an application that interacts with a blockchain or smart contracts instead of depending entirely on a centralized server.

PART 7 – QUICK REVISION BEFORE THE PRACTICAL

Term	One-line explanation
Blockchain	Decentralized digital ledger
Cryptocurrency	Digital currency secured using cryptography
Ethereum	Blockchain platform
Ether (ETH)	Native cryptocurrency of Ethereum
MetaMask	Cryptocurrency wallet
Public Address	Used to receive cryptocurrency
Private Key	Used to sign transactions
Recovery Phrase	Used to restore a wallet
Mainnet	Actual production network
Testnet	Network used for testing
Sepolia	Ethereum test network
Faucet	Provides free test ETH
Transaction	Operation performed on blockchain
Gas	Unit of computational work
Wei	Smallest unit of ETH
Gwei	Unit commonly used for gas prices
Remix IDE	Used to develop and deploy smart contracts
Etherscan	Used to explore blockchain transactions
Smart Contract	Program deployed on blockchain

PART 8 – FINAL PRACTICAL RESULT

Successfully created a cryptocurrency wallet using MetaMask, generated a unique wallet address, configured the Ethereum Sepolia Test Network and obtained free test ETH using a

faucet. The wallet was connected to Remix IDE for performing blockchain transactions and deploying smart contracts.